

Creation Date 07-May-2021

Revision Date 07-May-2021

Revision Number 1

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

Product Description: Thymol blue solution  
Cat No. : SP/3996/21

Unique Formula Identifier (UFI) **1FGX-K6GY-0X00-6TVW**

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### 1.3. Details of the supplier of the safety data sheet

Company **UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road, Loughborough,  
Leicestershire LE11 5RG, United Kingdom

**EU entity/business name**  
Acros Organics BVBA  
Janssen Pharmaceuticaaan 3a  
2440 Geel, Belgium

E-mail address [begel.sdsdesk@thermofisher.com](mailto:begel.sdsdesk@thermofisher.com)

### 1.4. Emergency telephone number

Tel: 01509 231166  
Chemtrec US: (800) 424-9300  
Chemtrec EU: 001 (202) 483-7616

Poison Centre - Emergency information services **Ireland** : National Poisons Information Centre (NPIC) - **01 809 2166** (8am-10pm, 7 days a week)  
**Malta** : +356 2395 2000  
**Cyprus** : +357 2240 5611

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

#### Physical hazards

Flammable liquids

Category 3 (H226)

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## Health hazards

Based on available data, the classification criteria are not met

## Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Warning

## Hazard Statements

H226 - Flammable liquid and vapor

## Precautionary Statements

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

## 2.3. Other hazards

No information available

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.2. Mixtures

| Component                                                                                                    | CAS-No     | EC-No.            | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                            |
|--------------------------------------------------------------------------------------------------------------|------------|-------------------|----------|--------------------------------------------------------------------------------------------------------------|
| Water                                                                                                        | 7732-18-5  | 231-791-2         | 81-82    | -                                                                                                            |
| Phenol,<br>4,4'-(3H-2,1-benzoxathiol-3-ylidene)bis[5-methyl-2-(1-methylethyl)-, S,S-dioxide, monosodium salt | 62625-21-2 | EEC No. 263-650-6 | <0.1     | -                                                                                                            |
| Ethyl alcohol                                                                                                | 64-17-5    | 200-578-6         | 17-18    | Flam. Liq. 2 (H225)<br>Eye Irrit. 2 (H319)                                                                   |
| Methyl alcohol                                                                                               | 67-56-1    | 200-659-6         | <0.99    | Flam. Liq. 2 (H225)<br>Acute Tox. 3 (H301)<br>Acute Tox. 3 (H311)<br>Acute Tox. 3 (H331)<br>STOT SE 1 (H370) |

| Component      | Specific concentration limits (SCL's) | M-Factor | Component notes |
|----------------|---------------------------------------|----------|-----------------|
| Ethyl alcohol  | Eye Irrit. 2 :: C>=50%                | -        | -               |
| Methyl alcohol | STOT SE 1 :: C>=10%                   | -        | -               |

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|  |                        |  |  |
|--|------------------------|--|--|
|  | STOT SE 2 :: 3%<=C<10% |  |  |
|--|------------------------|--|--|

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                  |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | If symptoms persist, call a physician.                                                                                                           |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                |
| <b>Ingestion</b>                          | Clean mouth with water and drink afterwards plenty of water.                                                                                     |
| <b>Inhalation</b>                         | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                     |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

### 4.2. Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

### 4.3. Indication of any immediate medical attention and special treatment needed

|                           |                                                 |
|---------------------------|-------------------------------------------------|
| <b>Notes to Physician</b> | Treat symptomatically. Symptoms may be delayed. |
|---------------------------|-------------------------------------------------|

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### Suitable Extinguishing Media

Water mist may be used to cool closed containers.

#### Extinguishing media which must not be used for safety reasons

No information available.

### 5.2. Special hazards arising from the substance or mixture

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

#### Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

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## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

### 6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system.

### 6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

#### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame.

**Technical Rules for Hazardous Substances (TRGS) 510 Storage Class (LGK)**  
(Germany)

Class 3

### 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### **Exposure limits**

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Third edition. Published 2018. **IRE** - 2018 Code of Practice for the Chemical Agents Regulations, Schedule 1. Published by the Health and Safety Authority

| Component     | The United Kingdom                            | European Union | Ireland               |
|---------------|-----------------------------------------------|----------------|-----------------------|
| Ethyl alcohol | TWA: 1000 ppm TWA; 1920 mg/m <sup>3</sup> TWA |                | STEL: 1000 ppm 15 min |

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|                |                                                                                                                 |                                                              |                                                                                                                              |
|----------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
|                | WEL - STEL: 3000 ppm<br>STEL: 5760 mg/m <sup>3</sup> STEL                                                       |                                                              |                                                                                                                              |
| Methyl alcohol | WEL - TWA: 200 ppm TWA;<br>266 mg/m <sup>3</sup> TWA<br>WEL - STEL: 250 ppm<br>STEL: 333 mg/m <sup>3</sup> STEL | TWA: 200 ppm 8 hr<br>TWA: 260 mg/m <sup>3</sup> 8 hr<br>Skin | TWA: 200 ppm 8 hr.<br>TWA: 260 mg/m <sup>3</sup> 8 hr.<br>STEL: 600 ppm 15 min<br>STEL: 780 mg/m <sup>3</sup> 15 min<br>Skin |

## Biological limit values

List source(s):

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                          | Acute effects local (Oral) | Acute effects systemic (Oral) | Chronic effects local (Oral) | Chronic effects systemic (Oral) |
|------------------------------------|----------------------------|-------------------------------|------------------------------|---------------------------------|
| Ethyl alcohol<br>64-17-5 ( 17-18 ) |                            | 87 mg/kg bw/d                 |                              |                                 |

| Component                           | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|-------------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Ethyl alcohol<br>64-17-5 ( 17-18 )  |                              |                                 |                                | DNEL = 343mg/kg<br>bw/day         |
| Methyl alcohol<br>67-56-1 ( <0.99 ) |                              | DNEL = 20mg/kg<br>bw/day        |                                | DNEL = 20mg/kg<br>bw/day          |

| Component                           | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|-------------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Ethyl alcohol<br>64-17-5 ( 17-18 )  | DNEL = 1900mg/m <sup>3</sup>     |                                     |                                    | DNEL = 950mg/m <sup>3</sup>           |
| Methyl alcohol<br>67-56-1 ( <0.99 ) | DNEL = 130mg/m <sup>3</sup>      | DNEL = 130mg/m <sup>3</sup>         | DNEL = 130mg/m <sup>3</sup>        | DNEL = 130mg/m <sup>3</sup>           |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                           | Fresh water     | Fresh water sediment           | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)          |
|-------------------------------------|-----------------|--------------------------------|--------------------|------------------------------------|-----------------------------|
| Ethyl alcohol<br>64-17-5 ( 17-18 )  | PNEC = 0.96mg/L | PNEC = 3.6mg/kg<br>sediment dw | PNEC = 2.75mg/L    | PNEC = 580mg/L                     | PNEC = 0.63mg/kg<br>soil dw |
| Methyl alcohol<br>67-56-1 ( <0.99 ) | PNEC = 20.8mg/L | PNEC = 77mg/kg<br>sediment dw  | PNEC = 1540mg/L    | PNEC = 100mg/L                     | PNEC = 100mg/kg<br>soil dw  |

| Component                           | Marine water    | Marine water sediment          | Marine water intermittent | Food chain                                         | Air |
|-------------------------------------|-----------------|--------------------------------|---------------------------|----------------------------------------------------|-----|
| Ethyl alcohol<br>64-17-5 ( 17-18 )  | PNEC = 0.79mg/L | PNEC = 2.9mg/kg<br>sediment dw |                           | PNEC = 0.38g/kg<br>food<br>PNEC = 0.72g/kg<br>food |     |
| Methyl alcohol<br>67-56-1 ( <0.99 ) | PNEC = 2.08mg/L | PNEC = 7.7mg/kg<br>sediment dw |                           |                                                    |     |

## 8.2. Exposure controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to

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control hazardous materials at source

## Personal protective equipment

**Eye Protection** Wear safety glasses with side shields (or goggles) (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Viton (R)      | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

**Skin and body protection** Long sleeved clothing.

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** low boiling organic solvent Type AX Brown conforming to EN371 or Organic gases and vapours filter Type A Brown conforming to EN14387

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141  
When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                  |                          |                                          |
|----------------------------------|--------------------------|------------------------------------------|
| <b>Physical State</b>            | Liquid                   |                                          |
| <b>Appearance</b>                |                          |                                          |
| <b>Odor</b>                      | No information available |                                          |
| <b>Odor Threshold</b>            | No data available        |                                          |
| <b>Melting Point/Range</b>       | No data available        |                                          |
| <b>Softening Point</b>           | No data available        |                                          |
| <b>Boiling Point/Range</b>       | No information available |                                          |
| <b>Flammability (liquid)</b>     | Flammable                | On basis of test data                    |
| <b>Flammability (solid,gas)</b>  | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>          | No data available        |                                          |
| <b>Flash Point</b>               | 32 °C / 89.6 °F          | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>  | No data available        |                                          |
| <b>Decomposition Temperature</b> | No data available        |                                          |
| <b>pH</b>                        | 6                        |                                          |
| <b>Viscosity</b>                 | No data available        |                                          |
| <b>Water Solubility</b>          | Soluble                  |                                          |

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|                                         |                          |             |
|-----------------------------------------|--------------------------|-------------|
| Solubility in other solvents            | No information available |             |
| Partition Coefficient (n-octanol/water) |                          |             |
| Component                               | log Pow                  |             |
| Ethyl alcohol                           | -0.32                    |             |
| Methyl alcohol                          | -0.74                    |             |
| Vapor Pressure                          | No data available        |             |
| Density / Specific Gravity              | 0.94                     |             |
| Bulk Density                            | Not applicable           | Liquid      |
| Vapor Density                           | No data available        | (Air = 1.0) |
| Particle characteristics                | Not applicable (liquid)  |             |

## 9.2. Other information

VOC Content(%) 18.2 explosive air/vapour mixtures possible

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

Hazardous Polymerization No information available.  
Hazardous Reactions None under normal processing.

### 10.4. Conditions to avoid

Keep away from open flames, hot surfaces and sources of ignition.

### 10.5. Incompatible materials

Strong oxidizing agents.

### 10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

|            |                                                                  |
|------------|------------------------------------------------------------------|
| Oral       | Based on available data, the classification criteria are not met |
| Dermal     | Based on available data, the classification criteria are not met |
| Inhalation | Based on available data, the classification criteria are not met |

#### Toxicology data for the components

| Component      | LD50 Oral                                                    | LD50 Dermal                   | LC50 Inhalation                                                   |
|----------------|--------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------|
| Water          | -                                                            | -                             | -                                                                 |
| Ethyl alcohol  | LD50 = 10470 mg/kg<br>OECD 401 (Rat)<br>3450 mg/kg ( Mouse ) | -                             | LC50 = 117-125 mg/l (4h)<br>OECD 403 (rat)<br>20000 ppm/10H (rat) |
| Methyl alcohol | LD50 > 1187 – 2769 mg/kg ( Rat )                             | LD50 = 17100 mg/kg ( Rabbit ) | LC50 = 128.2 mg/L ( Rat ) 4 h                                     |

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(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;

Respiratory No data available

Skin No data available

| Component                           | Test method                                                    | Test species | Study result    |
|-------------------------------------|----------------------------------------------------------------|--------------|-----------------|
| Ethyl alcohol<br>64-17-5 ( 17-18 )  | Mouse Ear Swelling Test (MEST)                                 | mouse        | non-sensitising |
|                                     | -----<br>OECD Test Guideline 429<br>Local Lymph Node Assay     | mouse        | non-sensitising |
| Methyl alcohol<br>67-56-1 ( <0.99 ) | OECD Test Guideline 406<br>Guinea Pig Maximisation Test (GPMT) | guinea pig   | non-sensitising |

(e) germ cell mutagenicity; No data available

| Component                          | Test method                                            | Test species          | Study result |
|------------------------------------|--------------------------------------------------------|-----------------------|--------------|
| Ethyl alcohol<br>64-17-5 ( 17-18 ) | AMES test<br>OECD Test Guideline 471                   | in vitro<br>Bacteria  | negative     |
|                                    | -----<br>Gene cell mutation<br>OECD Test Guideline 476 | in vitro<br>Mammalian | negative     |

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

| Component                           | Test method                      | Test species / Duration          | Study result              |
|-------------------------------------|----------------------------------|----------------------------------|---------------------------|
| Ethyl alcohol<br>64-17-5 ( 17-18 )  | OECD Test Guideline 416          | Oral / mouse<br>2 Generation     | NOAEL = 13.8 g/kg/day     |
|                                     | -----<br>OECD Test Guideline 414 | Inhalation / Rat                 | NOAEC =<br>16000 ppm      |
| Methyl alcohol<br>67-56-1 ( <0.99 ) | OECD Test Guideline 416          | Rat / Inhalation<br>2 Generation | NOAEC =<br>1.3 mg/l (air) |

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

## 11.2. Information on other hazards



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**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

#### Ecotoxicity effects

Contains a substance which is: Toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component      | Freshwater Fish                                            | Water Flea                                    | Freshwater Algae                           |
|----------------|------------------------------------------------------------|-----------------------------------------------|--------------------------------------------|
| Ethyl alcohol  | Fathead minnow (Pimephales promelas) LC50 = 14200 mg/l/96h | EC50 = 9268 mg/L/48h<br>EC50 = 10800 mg/L/24h | EC50 (72h) = 275 mg/l (Chlorella vulgaris) |
| Methyl alcohol | Pimephales promelas: LC50 > 10000 mg/L 96h                 | EC50 > 10000 mg/L 24h                         |                                            |

| Component      | Microtox                                                                                                    | M-Factor |
|----------------|-------------------------------------------------------------------------------------------------------------|----------|
| Ethyl alcohol  | Photobacterium phosphoreum: EC50 = 34634 mg/L/30 min<br>Photobacterium phosphoreum: EC50 = 35470 mg/L/5 min |          |
| Methyl alcohol | EC50 = 39000 mg/L 25 min<br>EC50 = 40000 mg/L 15 min<br>EC50 = 43000 mg/L 5 min                             |          |

### 12.2. Persistence and degradability

#### Persistence

Soluble in water, Persistence is unlikely, based on information available.

| Component                           | Degradability                  |
|-------------------------------------|--------------------------------|
| Ethyl alcohol<br>64-17-5 ( 17-18 )  | OECD 301E = 94%                |
| Methyl alcohol<br>67-56-1 ( <0.99 ) | DT50 ~ 17.2d<br>>94% after 20d |

#### Degradation in sewage treatment plant

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

### 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

| Component      | log Pow | Bioconcentration factor (BCF) |
|----------------|---------|-------------------------------|
| Ethyl alcohol  | -0.32   | No data available             |
| Methyl alcohol | -0.74   | <10                           |

### 12.4. Mobility in soil

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

### 12.5. Results of PBT and vPvB assessment

No data available for assessment.

### 12.6. Endocrine disrupting properties

#### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

### 12.7. Other adverse effects

#### Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

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## 13.1. Waste treatment methods

### Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

### European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

### Other Information

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations.

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

#### 14.1. UN number

UN1170

#### 14.2. UN proper shipping name

ETHANOL SOLUTION

#### 14.3. Transport hazard class(es)

3

#### 14.4. Packing group

III

### ADR

#### 14.1. UN number

UN1170

#### 14.2. UN proper shipping name

ETHANOL SOLUTION

#### 14.3. Transport hazard class(es)

3

#### 14.4. Packing group

III

### IATA

#### 14.1. UN number

UN1170

#### 14.2. UN proper shipping name

ETHANOL SOLUTION

#### 14.3. Transport hazard class(es)

3

#### 14.4. Packing group

III

#### 14.5. Environmental hazards

No hazards identified

#### 14.6. Special precautions for user

No special precautions required

#### 14.7. Maritime transport in bulk according to IMO instruments

Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

X = listed, Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), China (IECSC), Japan (ENCS), Australia (AICS), Korea (ECL).

| Component | EINECS    | ELINCS | NLP | TSCA | DSL | NDSL | PICCS | ENCS | IECSC | AICS | KECL         |
|-----------|-----------|--------|-----|------|-----|------|-------|------|-------|------|--------------|
| Water     | 231-791-2 | -      |     | X    | X   | -    | X     | X    | X     | X    | KE-3540<br>0 |
| Phenol,   | 263-650-6 | -      |     | X    | X   | -    | -     | -    | X     | -    | -            |

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|                                                                                                   |           |   |  |   |   |   |   |   |   |   |          |
|---------------------------------------------------------------------------------------------------|-----------|---|--|---|---|---|---|---|---|---|----------|
| 4,4'-(3H-2,1-benzoxathiol-3-ylidene)bis[5-methyl-2-(1-methylethyl)-, S,S-dioxide, monosodium salt |           |   |  |   |   |   |   |   |   |   |          |
| Ethyl alcohol                                                                                     | 200-578-6 | - |  | X | X | - | X | X | X | X | KE-13217 |
| Methyl alcohol                                                                                    | 200-659-6 | - |  | X | X | - | X | X | X | X | KE-23193 |

| Component      | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                                                           | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|----------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Methyl alcohol |                                                                     | Use restricted. See item 69.<br>(see <a href="http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=CELEX:32006R1907:EN:NOT">http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=CELEX:32006R1907:EN:NOT</a> for restriction details) |                                                                                                       |

| Component      | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|----------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Methyl alcohol | 500 tonne                                                                                 | 5000 tonne                                                                               |

**Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals**  
Not applicable

## National Regulations

### WGK Classification

Water endangering class = 1 (self classification)

| Component      | Germany - Water Classification (VwVwS) | Germany - TA-Luft Class |
|----------------|----------------------------------------|-------------------------|
| Ethyl alcohol  | WGK1                                   |                         |
| Methyl alcohol | WGK 2                                  |                         |

| Component      | France - INRS (Tables of occupational diseases)      |
|----------------|------------------------------------------------------|
| Ethyl alcohol  | Tableaux des maladies professionnelles (TMP) - RG 84 |
| Methyl alcohol | Tableaux des maladies professionnelles (TMP) - RG 84 |

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H225 - Highly flammable liquid and vapor  
H301 - Toxic if swallowed  
H311 - Toxic in contact with skin  
H319 - Causes serious eye irritation  
H331 - Toxic if inhaled  
H370 - Causes damage to organs

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

# SAFETY DATA SHEET

Thymol blue solution

Revision Date 07-May-2021

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**IECSC** - Chinese Inventory of Existing Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances  
**AICS** - Australian Inventory of Chemical Substances  
**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit  
**ACGIH** - American Conference of Governmental Industrial Hygienists  
**DNEL** - Derived No Effect Level  
**RPE** - Respiratory Protective Equipment  
**LC50** - Lethal Concentration 50%  
**NOEC** - No Observed Effect Concentration  
**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average  
**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)  
**LD50** - Lethal Dose 50%  
**EC50** - Effective Concentration 50%  
**POW** - Partition coefficient Octanol:Water  
**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road  
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code  
**OECD** - Organisation for Economic Co-operation and Development  
**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**ATE** - Acute Toxicity Estimate  
**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

|                              |                       |
|------------------------------|-----------------------|
| <b>Physical hazards</b>      | On basis of test data |
| <b>Health Hazards</b>        | Calculation method    |
| <b>Environmental hazards</b> | Calculation method    |

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

|                         |                  |
|-------------------------|------------------|
| <b>Creation Date</b>    | 07-May-2021      |
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**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006**

## Disclaimer

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**End of Safety Data Sheet**